

Multiple Choice (10 marks)

1. The proportion of genes shared by first cousins is on average _____.
 - (a) $1.0/2$
 - (b) $1.0/4$
 - (c) $1.0/8$
 - (d) $1.0/16$
2. Workers in which of the following jobs still have mandatory retirements once they reach a certain age?
 - (a) Cab drivers
 - (b) Fire fighters
 - (c) Insurance regulators
 - (d) Computer programmers
3. Which of the following is most strongly associated with loss of bone density?
 - (a) Genetic factors
 - (b) Lifestyle factors like not exercising
 - (c) Hormonal factors such as loss of estrogen
 - (d) Cohort factors
4. Intergenerational transmission to offspring can occur as a result of parental exposures to _____.
 - (a) war
 - (b) natural disasters
 - (c) hunger
 - (d) all of these
5. The surveillance testing strategy associated with the least selection bias is:
 - (a) Compulsory
 - (b) Unlinked anonymous
 - (c) Voluntary anonymous
 - (d) Voluntary confidential

True / False (10 marks)

Answer True or False

6. True or False: Needle aspiration is a method used to detect breast cancer that involves exposure to radiation.
7. True or False: A cross-sectional design allows researchers to analyze changes over time in both age and cohort effects.
8. True or False: Loss of smell is infrequent in adults aged 80 and older, affecting less than 40% of individuals in this age group.
9. True or False: Bile enters the digestive tract at the duodenum, where it aids in the emulsification of fats.
10. True or False: A pulse may be felt on a patient's face where the facial artery crosses the mandible.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the emotional impact of surprise visits from college grandchildren on their grandparents, focusing on feelings of love and the significance of seeking advice in their relationship.
12. Discuss the mechanism by which azidothymidine inhibits herpes viruses and analyze its effectiveness compared to other antiviral agents used in treatment.

End of Paper